Module 09 Lab - Ensemble Methods**Objective:** To learn how to use **ensemble methods**, which combine the predictions of several individual models to produce a final prediction that is more accurate and robust than any of the individual models alone.**In this lab, you will train and compare a single Decision Tree against a powerful ensemble method, the Random Forest.**

Part 1: The Wisdom of the Crowd

Concept: The core idea behind ensemble methods is the "wisdom of the crowd." If you ask one person a difficult question, they might be wrong. But if you ask a large, diverse group of people the same question and average their answers, the result is often surprisingly accurate.Ensemble methods apply this to machine learning. Instead of relying on a single model, we build a "committee" of models and have them vote on the final prediction.Two popular ensemble techniques:* **Bagging (Bootstrap Aggregating):** Trains multiple models (e.g., Decision Trees) on different random subsets of the training data. The **Random Forest** is the most famous example.* **Boosting:** Trains models sequentially, where each new model focuses on correcting the errors made by the previous one. Examples include Gradient Boosting and AdaBoost.

Part 2: SetupWe will use the Iris dataset to classify flower species.

```
import pandas as pd
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score

# Load and prepare data
iris = load_iris()
X = iris.data
y = iris.target
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)
```

Part 3: A Single Decision TreeA single Decision Tree is prone to overfitting. It can create very specific rules that match the training data perfectly but don't generalize well.**Your Task:** Train a single `DecisionTreeClassifier` and evaluate its accuracy.

```
# --- ENTER YOUR CODE HERE ---
# 1. Create and train a single Decision Tree
single_tree = DecisionTreeClassifier(random_state=42)
single_tree.fit(X_train, y_train)
```

```
# 2. Make predictions and evaluate its accuracy
y_pred_tree = single_tree.predict(X_test)
accuracy_tree = accuracy_score(y_test, y_pred_tree)
print(f"Accuracy of a single Decision Tree: {accuracy_tree:.2%}")
```

```
Accuracy of a single Decision Tree: 100.00%
```

Part 4: The Random Forest Ensemble

Concept: A Random Forest is an ensemble of many Decision Trees. It builds hundreds of trees, each trained on a random subset of the data and using a random subset of the features. To make a prediction, it gets a vote from every tree in the "forest" and chooses the class that gets the most votes.This process of averaging the votes of many diverse trees makes the Random Forest much more robust and less prone to overfitting than a single tree. **Your Task:** Train a `RandomForestClassifier` and compare its accuracy to the single tree.

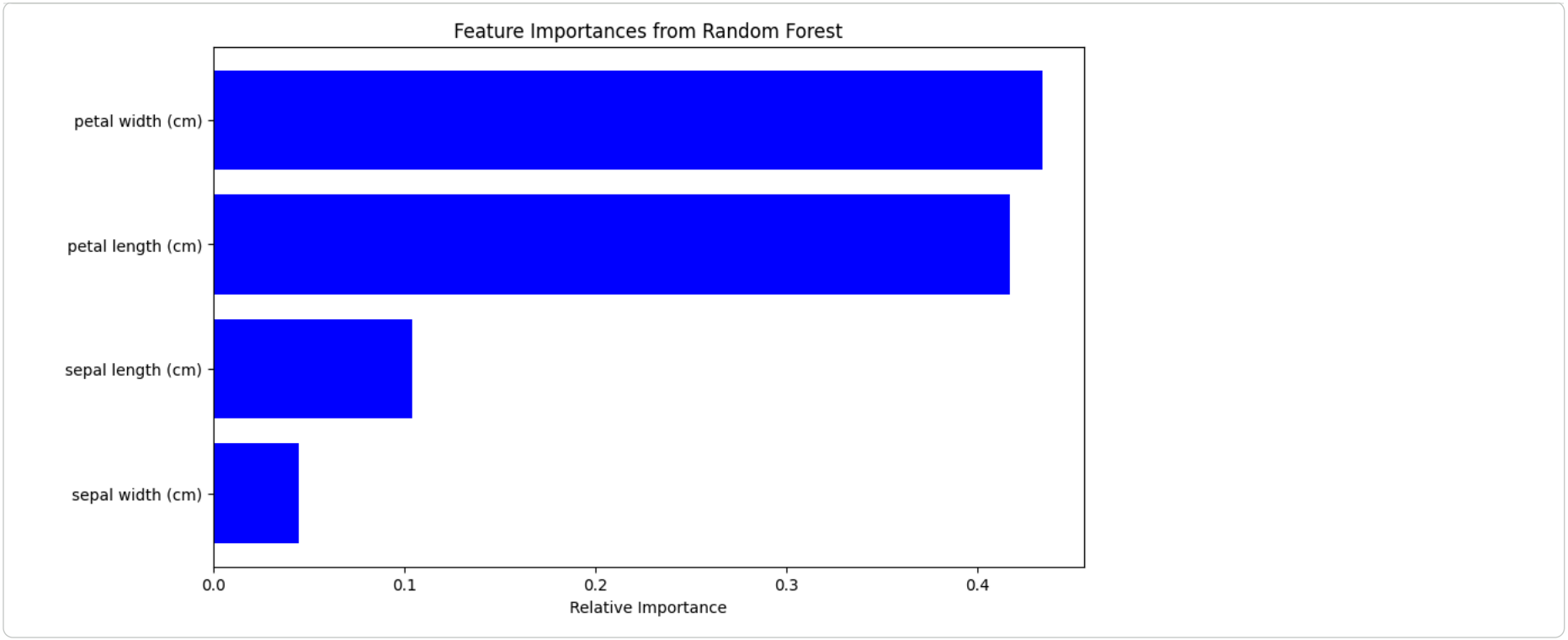
```
# --- ENTER YOUR CODE HERE ---
# 1. Create and train a Random Forest`n_estimators` is the number of trees in the forest.
random_forest = RandomForestClassifier(n_estimators=100, random_state=42)
random_forest.fit(X_train, y_train)

# 2. Make predictions and evaluate its accuracy
y_pred_rf = random_forest.predict(X_test)
accuracy_rf = accuracy_score(y_test, y_pred_rf)
print(f"Accuracy of the Random Forest: {accuracy_rf:.2%}")
```

```
Accuracy of the Random Forest: 100.00%
```

Part 5: Feature Importance**Concept:** A great benefit of tree-based ensembles like Random Forest is that they can tell you which features were most important for making accurate predictions.**Your Task:** Extract and plot the feature importances from your trained Random Forest model.

```
import matplotlib.pyplot as plt
import numpy as np
# --- ENTER YOUR CODE HERE ---
# 1. Get the feature importances from the trained Random Forest model
importances = random_forest.feature_importances_
# 2. Create a bar plot to visualize them
feature_names = iris.feature_names
indices = np.argsort(importances)
plt.figure(figsize=(10, 6))
plt.title("Feature Importances from Random Forest")
plt.barh(range(len(indices)), importances[indices], color='b', align='center')
plt.yticks(range(len(indices)), [feature_names[i] for i in indices])
plt.xlabel('Relative Importance')
plt.show()
```



Knowledge Check

Instructions: Answer the following questions in this markdown cell.**1.** In your own words, what is the main idea behind ensemble methods? Why is a "committee" of models often better than a single expert model? **2.** Which model performed better in this lab, the single Decision Tree or the Random Forest? Is this what you expected? Why? **3.** Looking at the feature importance plot, which two features were most important for classifying the iris flowers? ** Does this match what you might expect from your own knowledge of flowers? **ANSWERS 1.)** An ensemble method combines several models instead of trusting just one. Each model makes its own prediction, and the group votes (or averages) to get the final answer. A "committee" works better than a single expert because each model makes different mistakes, and those mistakes cancel each other out, which reduces the chance of error. **2.)** Both models scored 100%, so it was a tie. I did not expect that, because a Random Forest uses many trees voting together and usually beats a single tree. They tied because the Iris dataset is small and easy to separate, so even one Decision Tree could classify every flower correctly. **3.)** The two most important features were petal width (0.43) and petal length (0.42). They were the easiest way to tell the flowers apart. This matches what I'd expect, because petal size looks very different between iris types while sepals look more alike.